DHAKA UNIVERSITY OF ENGINEERING & TECHNOLOGY, GAZIPUR
B.Sc. Engineering 4th year 1st Semester **(Regular)** Examination, 2024
Department of Electrical and Electronic Engineering
Course Title: **Telecommunication-II**
Course No.: **EEE 4201/EEE-4203**

Full Marks: 210

Time: 3 hours

Instructions: (a) Answer **any three questions** from each section.
(b) Use **separate script** for each section.
(c) Figures in the right margin indicate full marks.

SECTION-A

1. (a) What is line coding? Why is it important to minimize DC components in line coding, and which coding schemes help achieve this? (10) CO1

(b) In a joint TDM and FDM multiplexing system, 20 sources are multiplexed in each TDM block. The bandwidth of each source is 4 kHz. The Nyquist sampling rate is used for sampling the sources and each sample of the sources is converted to 8 bits by binary encoding. Interleaved unit of the TDM blocks is a sample of 8 bits and 1 bit are used for the synchronization of each frame. An FDM block is used to multiplex the outputs of 4 TDM blocks with ASK modulation of 4 different carrier frequencies with a guard band of 2 kHz. Determine: (15) CO2

(i) the data rate of each TDM output and
(ii) the bandwidth of the FDM output signal.

(c) The waveform shown in Fig. 1(c) represents the Multiline Transmission: MLT-3 encoding of a data stream. Decode the data stream. (10) CO2

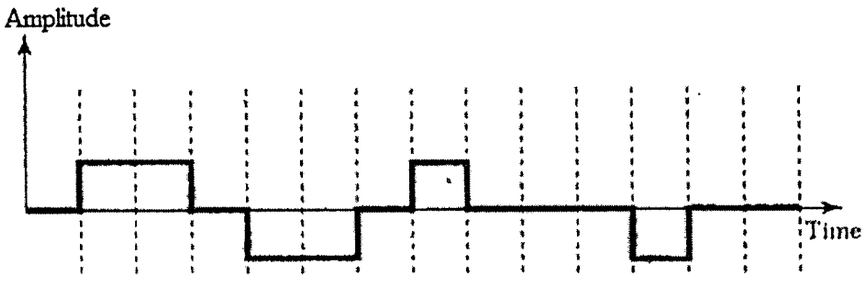

Fig. 1(c)

2. (a) Design a PSK transmitter capable of transmitting three bits per signal element. Also calculate the amplitudes and phases of the resulting PSK modulated signal for all possible input bit combinations. (12) CO2

(b) Sketch and explain the coherent detection of a BFSK signal, providing the necessary mathematical derivation. (13) CO1

(c) A binary baseband signal is passed through a raised cosine roll-off filter with a 40% roll-off factor and is then modulated onto a carrier. The data rate is 128 kbit/s. Evaluate: (10) CO2

(i) the absolute bandwidth of the resulting OOK signal.
(ii) the approximate bandwidth of the resulting FSK signal when the mark frequency is 250 kHz and the space frequency is 260 kHz.

3. (a) How does the M-ary modulation scheme differ from the binary modulation scheme? What is the significance of M-ary modulation schemes in digital communication systems? (10) CO1

(b) Draw the block diagram of an 8-QAM transmitter and receiver. Explain how the modulation and demodulation work with a specific input bit sequence. (15) CO1

(c) For an 8-PSK system, operating with an information bit rate of 24 kbps, determine: (10) CO2

(i) baud
(ii) minimum bandwidth and
(iii) bandwidth efficiency.

4. (a) What is thermal noise in a communication system? Derive the equation of equivalent thermal noise voltage and current with necessary sketches. (10) CO1

4. (b) An information packet contains 200 bits. This packet is transmitted over a communications channel where the probability of error for each bit is 10^{-3}. Suppose the packet includes an error-correcting code that can correct up to three errors located anywhere in the packet. What is the probability that a particular packet is received in error in this case? (10) CO2

(c) A network system consists of four computers, two routers, and five LANs. Each computer is connected to exactly one LAN, and each router is connected to three LANs. The network connections are as follows: Router 1 is connected to LAN 1, LAN 2, and LAN 4, while Router 2 is connected to LAN 2, LAN 3, and LAN 5. Computers 1, 2, 3, and 4 are connected to LAN 1, LAN 4, LAN 5, and LAN 3, respectively. Each device has a unique physical and logical address for each connection. The physical and logical addresses of the computers are given in the table below: (15) CO2

Device	Computer 1	Computer 2	Computer 3	Computer 4
Physical Address	33	57	65	80
Logical Address	G	Q	L	W

You have to send a data packet from computer 2 to computer 4. Draw the network connection diagram, showing the data packet with physical and logical addresses at each hop. Assume the router addresses for each connection as per your convenience.

SECTION-B

5. (a) Define critical propagation angle and total internal reflection. Briefly explain the different losses of optical fiber. (10) CO1

(b) A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.52 and a cladding refractive index of 1.50. Determine: (12) CO2

(i) The NA (numerical aperture) and critical propagation angles for the fiber.

(ii) The critical angle at core cladding interface, and

(iii) The acceptance angle in air for the fiber.

(c) Briefly explain dispersion in an optical fiber. Express the compensation of modal dispersion. (13) CO1

6. (a) Define attenuation for optical fiber communication. The mean optical power launched into an optical fiber link is 1.55 mW and the fiber has an attenuation of 0.25 dB/km. Determine the maximum possible link length without repeater when minimum optical power required at the detector is 2.55 μW. (11) CO2

(b) Define responsivity and quantum efficiency. Derive the relationship between responsivity and quantum efficiency, and provide the corresponding expression. (12) CO1

(c) A p-n photodiode has a quantum efficiency of 75% at a wavelength of 0.80 μm. Calculate: (i) its responsivity. (12) CO2

(ii) the received optical power if photo current is 20 μA.

(iii) the corresponding number of received photons.

7. (a) Define cellular concept. Why is the hexagonal shape widely used? (08) CO1

(b) Define cell splitting and cell sectoring. Derive the relationship between co-channel cell distance, D and the cluster size, N and provide the corresponding expression. (15) CO1

(c) Write short notes on (i) JPGG and (ii) MPGA. (12) CO1

8. (a) Describe the various generation, for instant 2G, 3G, 4G and 5G of cellular system. (13) CO1

(b) Define Radar. Briefly explain the working principle and application of Radar. (12) CO1

(c) Draw the block diagram of adaptive equalizer in the receiver and briefly describe each of them. (10) CO1

-THE END-

DHAKA UNIVERSITY OF ENGINEERING & TECHNOLOGY, GAZIPUR

B.Sc. Engineering 4th year 2nd Semester **(Regular)** Examination, 2023

Department of Electrical and Electronic Engineering

Course Title: **Telecommunication-II**

Course No.: **EEE-4203**

Time: 3 hours

Full Marks: 210

Instructions: (a) Answer **any three questions** from each section.
(b) Use **separate script** for each section.
(c) All questions are of **equal** value.
(d) Figures in the right margin indicate full marks.

SECTION-A

1. (a) Explain the process of generating a BPSK signal. Can such a signal be detected using non-coherent techniques? With the appropriate expressions and waveforms, illustrate each stage of the detection process. (CO1) (12)

   (b) Design and draw a DPSK transmitter and receiver, where a '1' indicates a transition. Justify your design by specifying the appropriate logic gate at the transmitter and the comparator logic at the receiver. Additionally, for the original data stream [1 0 1 0 0 1 1 0], determine (i) the differentially encoded bits, (ii) the transmitted phases, and (iii) the detected bits. (CO2) (13)

   (c) Digital information is to be transmitted by carrier modulation through an additive gaussian noise channel with bandwidth of 100 kHz and $N_0 = 10^{-10}\ W/Hz$. Determine the maximum rate that can be transmitted through the channel for four phase PSK, binary FSK and four-frequency orthogonal FSK, which is detected noncoherently. (CO2) (10)

2. (a) Describe the process of generating signals in an 8-PSK modulation scheme. (CO1) (12)

   (b) Derive the expression for the average probability of bit error for a coherent binary BPSK transmission system. (CO1) (13)

   (c) Assume that a 64-QAM signal is used to send data at a rate of 100 Mbits/s over a satellite transponder. The transponder has a bandwidth of 30 MHz. If the satellite signal is equalized to have an equivalent raised cosine filter characteristic, what is the rolloff factor r required? Could a rolloff factor r be found so that a 180 Mbit/s data rate could be supported? (CO2) (10)

3. (a) Briefly discuss the different types of radar systems. Additionally, explain how radar is utilized in Air Traffic Control systems. (CO1) (13)

   (b) Calculate the maximum range of a radar system with an operating frequency of 12 GHz. The radar transmits a peak power of 300 kW, and the receiving antenna has an effective aperture of 5 m². The target has a radar cross section area (RCS) of 25 m², and the minimum detectable signal power is 1 nW. (CO2) (10)

   (c) Explain the function of a satellite transponder using a detailed diagram. (CO1) (12)

4. (a) Explain the need for satellite launching methods. Describe any one method of satellite launching. (CO1) (12)

   (b) What is meant by a protocol in networking? Describe in detail the function of each layer in the OSI model. (CO1) (13)

   (c) Define data communication. Describe various LAN topologies with the necessary diagram. (CO1) (10)

SECTION-B

5. (a) Define acceptance angle and numerical aperture (NA). Show that NA depends not only on the core or cladding refractive index but on their difference. (CO1) (10)

   (b) Define attenuation in an optical fiber. List three major causes of attenuation in an optical fiber and explain their mechanism. (CO1) (13)

   (c) A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.52 and a cladding refractive index of 1.48. Determine: (CO2) (12)

      (i) The critical propagation angles.

      (ii) The critical angle at core cladding interface, and

      (iii) The acceptance angle in air for the fiber.

6. (a) Describe polarization mode dispersion in an optical fiber. Express the compensation of such dispersion. (CO1) (11)

   (b) Describe the main reasons to use LASER Diode (LD) instead of LED as a best light source for long-haul fiber optic links. Draw the stimulated emission, light amplification, and positive feed-back process in case of LD. (CO1) (12)

   (c) A p-n photodiode has a quantum efficiency of 75% at a wavelength of 0.87 μm. Calculate: (CO2) (12)

      (i) Its responsivity.

      (ii) The received optical power if photo current is 16 μA.

      (iii) The corresponding number of received photons.

7. (a) Describe the different types of handovers and explain how they work. (CO1) (13)

   (b) Elucidate frequency reuse concept of a cellular network. Show that, for a hexagonal geometry, the co-channel reuse ratio $Q = \sqrt{3N}$ where $N = i^2 + ij + j^2$. (CO1) (10)

   (c) If a total of 30 MHz of bandwidth is allocated to a particular FDD cellular telephone system which uses a 50 KHz simplex channels to provide full duplex voice and control channels, compute the number of channels available per cell if a system uses (a) 4-cell reuse (b) 7-cell reuse (c) 12-cell reuse. If 1 MHz of the allocated spectrum is dedicated to control channels, determine an equitable distribution of control channels and voice channels in each cell for each of the three systems. (CO2) (12)

8. (a) Draw the block diagram of adaptive equalizer in the receiver and briefly describe each of them. (CO1) (12)

   (b) Briefly describe the TDMA, FDMA and CDMA in wireless communication system. (CO1) (10)

   (c) Explicate the structure of a GSM network and describe each component with their role in the operation of GSM system. (CO1) (13)

-THE END-

DHAKA UNIVERSITY OF ENGINEERING & TECHNOLOGY, GAZIPUR

B.Sc. Engineering 4th year 2nd Semester **(Regular)** Examination, 2021

Department of Electrical and Electronic Engineering

Course Title: **Telecommunication-II**

Course No.: **EEE-4203**

Time: 3 hours | Full Marks: 210

Instructions: (a) Answer **any three questions** from each section.
(b) Use **separate script** for each section.
(c) All questions are of **equal** value.
(d) Figures in the right margin indicate full marks.

SECTION-A

1. (a) Briefly explain the relative advantages and disadvantages of analog and digital communication systems. (10) CO1

   (b) Consider an ASK-based digital communication system that transmits two different signals, $A_0 cos(\omega_c t)$ for bit 0 and $A_1 cos(\omega_c t)$ for bit 1. Draw a block diagram of the transmitter and receiver of the system, and provide the expressions for the signals at each point of the transmitter and receiver. (13) CO2

   (c) What is meant by MSK modulation scheme? Demonstrate that the minimum tone spacing for noncoherently detected orthogonal FSK signaling, where symbols are equiprobable, is given by:

   $$f_1 - f_2 = \frac{1}{T}$$

   Where symbols have their usual meaning. (12) CO1

2. (a) Mention the advantages of M-ary modulation scheme over the binary modulation. (08) CO1

   (b) Explain the generation and both the coherent & non-coherent detection system of BFSK modulation scheme. (15) CO1

   (c) Bangladesh Satellite Company Limited (BSCL) uses an AWGN channel with 6 kHz bandwidth and the noise power spectral density $n_0/2 = 10^{-12}\ W/Hz$. The signal power is required at the receiver is 0.2 mW. Find the capacity of the channel. If the above channel is to be used to transmit 100 kbps of data determine the minimum BW required of the channel of same noise power spectral density characteristics. (12) CO2

3. (a) Drive the radar range equation and from this equation, comment on the relationship between the maximum range of radar (R_{max}) and operating wavelength (λ). (13) CO1

   (b) Determine the maximum range of a radar system, provided that the peak transmitted power is 250 kW, minimum detectable signal power is 10 pW, the radar cross section (RCS) of the target is 20 m^2, the gain of the transmitting antenna is 4000, and the effective aperture of the receiver antenna is 4.5 m^2. (10) CO2

   (c) What are the main tasks of an earth station? Describe with block diagram how the terrestrial components communicate through a satellite. (12) CO1

4. (a) Explain the operation of a Doppler radar. (12) CO1

   (b) What is meant by protocol? Describe the function of each layer of TCP/IP model. (13) CO1

   (c) Briefly explain the frame transmission technique in a ring LAN. (10) CO1

SECTION-B

5. (a) Define acceptance angle and numerical aperture (NA). Show that NA depends not only on the core or cladding refractive index but on their difference. (10) CO1

(b) A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.50 and a cladding refractive index of 1.47. Determine:

(i) The critical propagation angles.

(ii) The critical angle at core cladding interface, and

(iii) The acceptance angle in air for the fiber. (12) CO2

(c) Illustrate the block diagram of fiber-optic communication system and describe the functions of each component. (13) CO1

6. (a) Explain polarization mode dispersion in an optical fiber. Express the compensation of such dispersion. (11) CO1

(b) Define attenuation for optical fiber communication. The mean optical power launched into an optical fiber link is 1.55 mW and the fiber has an attenuation of 0.25 dB/km. Determine the maximum possible link length without repeater when minimum optical power required at the detector is 2.55 μW. (12) CO2

(c) A p-n photodiode has a quantum efficiency of 70% at a wavelength of 0.85 μm. Calculate: (i) its responsivity.

(ii) the received optical power if photo current is 15 μA.

(iii) the corresponding number of received photons. (12) CO2

7. (a) Mention the classification of handover. Briefly explain the handover process. (13) CO1

(b) Explain frequency reuse concept of a cellular network. Show that, for a hexagonal geometry, the co-channel reuse ratio $Q = \sqrt{3N}$ where $N = i^2 + ij + j^2$. (10) CO1

(c) If a total of 35 MHz of bandwidth is allocated to a particular FDD cellular telephone system which uses a 55 KHz simplex channels to provide full duplex voice and control channels, compute the number of channels available per cell if a system uses (a) 4-cell reuse (b) 7-cell reuse (c) 12-cell reuse. If 1 MHz of the allocated spectrum is dedicated to control channels, determine an equitable distribution of control channels and voice channels in each cell for each of the three systems. (12) CO2

8. (a) Explain the structure of a GSM network and describe each component with their role in the operation of GSM system. (13) CO1

(b) Differentiate among the TDMA, FDMA and CDMA in wireless communication system. (12) CO1

(c) Describe the various generation, for instant 2G, 3G, 4G and 5G of cellular system. (13) CO1

-THE END-

DHAKA UNIVERSITY OF ENGINEERING & TECHNOLOGY, GAZIPUR

B.Sc. Engineering 4th year 2nd Semester Examination, 2019

Department of Electrical and Electronic Engineering

Course Title: **Telecommunication-II**

Course No.: **EEE-4203**

Time: 2 hours | Full Marks: 150

Instructions: (a) Answer **any three questions** from each section.
(b) Use **separate script** for each section.
(c) All questions are of **equal** value.
(d) Figures in the right margin indicate full marks.

SECTION-A

1. (a) We have a baseband channel with a 2-MHz bandwidth. What is the data rate for this channel if we use one of the following line coding schemes?
   (i) NRZ-L
   (ii) Manchester
   (iii) Differential Manchester
   (iv) NRZ-I (12)

   (b) The binary sequence "last two digits of your student ID" is applied to an BASK modulator. The bit duration is 1 μs, and sinusoidal carrier wave used to represent symbol 1 has a frequency equal to "last two digits of your student ID in decimal" MHz.
   (i) Find the transmission bandwidth of the transmitted signal
   (ii) Plot the waveform of the transmitted BASK signal. (13)

2. (a) The telephone line has 8kHz bandwidth. What is the maximum number of bits we can send using each of the following techniques? Let d = 0
   (i) BASK (ii) QPSK (iii) 64-QAM (iv) 128-QAM (10)

   (b) A BFSK signal with discontinuous phase is generated by using the incoming binary data stream to switch between two independent sinusoidal oscillators:

$$\text{Symbol 1: } \sqrt{\frac{2E_b}{T_b}} Cos(2\pi f_1 t + \theta_1)$$

$$\text{Symbol 0: } \sqrt{\frac{2E_b}{T_b}} Cos(2\pi f_2 t + \theta_2)$$

   (i) Show that this BFSK signal may be expressed as the sum of two independent BASK signals.
   (ii) For θ_1 = (last two digits of your student ID)°
   and θ_2 = (middle two digits of your student ID)°, plot the BFSK waveform for the binary data stream "01101001" and compare your result to continuous-phase BFSK. (15)

3. (a) For each of the following four networks, discuss the consequences if a connection fails:
   (i) Six devices arranged in a bus topology
   (ii) Four devices arranged in a ring topology
   (iii) Five devices arranged in a mesh topology
   (iv) Seven devices arranged in a star topology (not counting the hub). (10)

   (b) What is the purpose of a reference model in data communication system? How is the OSI reference model different from the TCP/IP reference model? Mention the responsibilities of the each layer of the OSI model. (15)

4. (a) What is the number of bits per baud for the following techniques?
   (i) ASK with 8 different amplitudes
   (ii) FSK with 16 different frequencies
   (iii) PSK with different phases
   (iv) QAM with a constellation of 256 points. (12)

   (b) Draw the OQPSK signal components for the data "1100011001":
   (i) Modulating signal for in-phase component
   (ii) Modulated waveform of in-phase component
   (iii) Modulating signal for quadrature component
   (iv) Modulated waveform of quadrature component
   (v) OQPSK waveform. (13)

SECTION-B

5. (a) Explain chromatic dispersion and polarization mode dispersion in an optical fiber. How can such dispersion be compensated? (15)

   (b) A silica optical fiber has n_1 = 1.78 and n_2 = 1.76. Find (i) the critical propagation angle, (ii) the acceptance angle and (iii) the numerical aperture. How much will a light pulse spread after travelling along 6 km of this step-index fiber. (10)

6. (a) Compare the characteristics of LED and LD as optical source. (05)

   (b) Define quantum efficiency and responsitivity. Show that at 1248 nm wavelength, responsitivity and quantum efficiency is equal. (10)

   (c) When the mean optical power lunched into an 8km length of fiber is 125μW, the mean optical power at the fiber output is 4μW. Determine:

   (i) The overall signal attenuation or loss in decibels through the fiber assuming there are no connectors or splices;

   (ii) The signal attenuation per kilometer for the fiber;

   (iii)The overall signal attenuation for a 10 km optical link using the same fiber with splices at 1 km intervals, each giving an attenuation of 1 dB;

   (iv)The numerical input/output power ratio in (iii). (10)

7. (a) Define cell and cluster. Why is the shape of the cell chosen as hexagonal? (08)

   (b) Explain how the capacity of cellular system is enhanced by "cell-splitting" and "sectoring": (08)

   (c) What is meant by equalization and diversity techniques for communication systems? (09)

8. (a) A p-n photo diode has a quantum efficiency of 60% at a wavelength of 0.9μm. Calculate:

   (i) its responsitivity at 0.9μm

   (ii) the received optical power if mean photo current is $10^{-6}A$

   (iii) the corresponding number of received photons at this wavelength. (13)

   (b) With necessary diagram explain the working principle of continuous-wave (C.W.) radar. (12)

-THE END-

DHAKA UNIVERSITY OF ENGINEERING & TECHNOLOGY, GAZIPUR

B.Sc. Engineering 4th year 2nd Semester (**Special**) Examination, 2017

Department of Electrical and Electronic Engineering

Course Title: **Telecommunication-II**

Course No.: **EEE-4203**

Time: 3 hours

Full Marks: 210

Instructions: (a) Answer **any three questions** from each section.
(b) Use **separate script** for each section.
(c) All questions are of **equal** value.
(d) Figures in the right margin indicate full marks.

SECTION-A

1. (a) Briefly explain the relative advantages and disadvantages of analog and digital communication system. (08)
   (b) Mention the advantages of coherent modulation scheme over non-coherent scheme. (08)
   (c) Describe the modulator and demodulator circuit of FSK modulation scheme. (14)
   (d) Why ASK is more affected by noise? (05)

2. (a) Derive the signal-space diagram of QPSK and 16-QAM signal. (10)
   (b) Describe QAM modulation process using the constellation diagram of 4-QAM. (12)
   (c) Draw the block diagram of DPSK transmitter and receiver and explain each block briefly. (13)

3. (a) Explain the operation of Doppler radar. (13)
   (b) Derive the free-space radar range equation. Explain the factors on which radar range depend. (12)
   (c) Describe the satellite launching method in a geo-stationary orbit. (10)

4. (a) Explain the different sub-station of satellite earth station. (15)
   (b) Describe different LAN topologies with necessary diagram. (10)
   (c) Briefly describe different layers of TCP/IP model. (10)

SECTION-B

5. (a) Draw the block diagram of an optical fiber communication system and explain each block briefly. (10)
   (b) Explain the construction of an optical fiber. (07)
   (c) Define attenuation. Enumerate main reasons of attenuation and caused in an optical communication system. (12)
   (d) A silica fiber has $n_1 = 1.4513$ and $n_2 = 1.4468$. Find (i) critical propagation angle, (ii) acceptance angle and (iii) numerical aperture. (06)

6. (a) Express the pulse-spreading time caused by modal dispersion in terms of relative refractive index difference. (12)
   (b) Compare the characteristics of LED and LD as optical source. (08)
   (c) What is meant by responsibility of photo diode? Derive, $R = \frac{\eta e \lambda}{hc}$, where symbols have their usual meanings. (15)

7. (a) Define cell and cluster. Why is the shape of a cell chosen as hexagonal? (08)
   (b) Explain the architecture of a GSM cellular network. (15)
   (c) Explain the handover process in a GSM cellular network. (12)

8. (a) Explain the difference among FDMA, TDMA and CDMA. (09)
   (b) Explain the evaluation of data services for different generation of mobile network. (11)
   (c) What do you mean by diversity technique in communication system? Describe different diversity technique. (15)

-THE END-

DHAKA UNIVERSITY OF ENGINEERING & TECHNOLOGY, GAZIPUR
B.Sc. Engineering 4th year 2nd Semester (**Special**) Examination, 2020
Department of Electrical and Electronic Engineering
Course Title: **Telecommunication - II**
Course No.: **EEE-4203**

Time: 3 hours | Full Marks: 210

Instructions: (a) Answer **any three questions** from each section.
(b) Use **separate script** for each section.
(c) All questions are of **equal** value.
(d) Figures in the right margin indicate full marks.
(e) All the symbols have their usual meanings.

SECTION-A

1. (a) Explain ASK, FSK and PSK modulation schemes with their applications. (12) CO1
   (b) Derive the expression for average probability of symbol error for coherent binary FSK system. (13) CO2
   (c) Why is coherent detection required for BPSK modulation? Draw the wave-shapes for modulated signal using BASK, BFSK, BPSK and QPSK modulation schemes to transmit a bit sequence '01101001'. (10) CO2

2. (a) Briefly explain the generation and detection of differential PSK (DPSK) signal. (10) CO1
   (b) Derive the signal-space diagram of 8-PSK and 16-QAM signal. (12) CO2
   (c) The binary sequence 01101000 is applied to a QPSK modulator. This sequence is divided in to two other sequences, consisting of odd and even numbered bits of the input sequence, which are basically representing the in-phase and quadrature components, respectively. Determine the final QPSK waveform. (13) CO2

3. (a) Describe the operation of a Doppler radar and mention its application. (17) CO2
   (b) What is speech coding? What are the objectives of speech coding? (08) CO1
   (c) What is meant by MSK? Derive the expression of the phase in MSK for symbol 0 and 1. (10) CO1

4. (a) Using the block diagram of earth station and transponder describe the working principle of a satellite. (15) CO1
   (b) Describe the launching method of satellite. (10) CO1
   (c) Why is the uplink frequency different from downlink frequency in a satellite communication system? (10) CO1

SECTION-B

5. (a) List the advantages and disadvantages of optical fiber. Draw the basic block diagram of optical fiber communication system and the inner parts (construction) of optical fiber. (12) CO1
   (b) A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.50 and a cladding refractive index of 1.48. Determine: (10) CO1
      (i) The critical propagation angles.
      (ii) The critical angle at core cladding interface, and
      (iii) The acceptance angle in air for the fiber.
   (c) Describe the main reasons to use LD instead of LED as a best light source for long-haul fiber optic links. Draw the stimulated emission, light amplification and positive feed-back process in case of LD. (13) CO2

6. (a) Define acceptance angle and numerical aperture (NA). Show that NA depends not only on the core or cladding refractive index but on their difference. (12) CO1
   (b) Define attenuation for optical fiber communication. The mean optical power launched into an optical fiber link is 1.6 mW and the fiber has an attenuation of 0.26 dB/km. Determine the maximum possible link length without repeater when minimum optical power required at the detector is 2.5 μW. (12) CO2

(c) The velocity of light in the core of a step index fiber is $2.05 \times 10^8\ m/s$ and the critical angle at the core-cladding interface is 70°. Determine NA, acceptance angle and critical propagation angle. (11) CO2

7. (a) Mention the classification of handover. Briefly explain the handover process. (13) CO1

(b) Explain frequency reuse concept of a cellular network. Show that, for a hexagonal geometry, the co-channel reuse ratio $Q = \sqrt{3N}$. (10) CO1

(c) If a total of 35 MHz of bandwidth is allocated to a particular FDD cellular telephone system which uses a 50 KHz simplex channels to provide full duplex voice and control channels, compute the number of channels available per cell if a system uses (a) 4-cell reuse (b) 7-cell reuse (c) 12-cell reuse. If 1 MHz of the allocated spectrum is dedicated to control channels, determine an equitable distribution of control channels and voice channels in each cell for each of the three systems. (12) CO2

8. (a) Draw and explain a simplified communication system using an adaptive equalizer in the receiver. (10) CO1

(b) Differentiate among the TDMA, FDMA and CDMA in wireless communication system. (12) CO1

(c) Compare among various generations of cellular system. (13) CO1

-THE END-

DHAKA UNIVERSITY OF ENGINEERING & TECHNOLOGY, GAZIPUR

B.Sc. Engineering 4th year 2nd Semester Examination, 2019

Department of Electrical and Electronic Engineering

Course Title: **Telecommunication-II**

Course No.: **EEE-4203**

Time: 2 hours | Full Marks: 150

Instructions: (a) Answer **any three questions** from each section.
(b) Use **separate script** for each section.
(c) All questions are of **equal** value.
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SECTION-A

1. (a) We have a baseband channel with a 2-MHz bandwidth. What is the data rate for this channel if we use one of the following line coding schemes?
   (i) NRZ-L
   (ii) Manchester
   (iii) Differential Manchester
   (iv) NRZ-I (12)

   (b) The binary sequence "last two digits of your student ID" is applied to an BASK modulator. The bit duration is 1 μs, and sinusoidal carrier wave used to represent symbol 1 has a frequency equal to "last two digits of your student ID in decimal" MHz.
   (i) Find the transmission bandwidth of the transmitted signal
   (ii) Plot the waveform of the transmitted BASK signal. (13)

2. (a) The telephone line has 8kHz bandwidth. What is the maximum number of bits we can send using each of the following techniques? Let d = 0
   (i) BASK (ii) QPSK (iii) 64-QAM (iv) 128-QAM (10)

   (b) A BFSK signal with discontinuous phase is generated by using the incoming binary data stream to switch between two independent sinusoidal oscillators:

$$\text{Symbol 1: } \sqrt{\frac{2E_b}{T_b}} Cos(2\pi f_1 t + \theta_1)$$

$$\text{Symbol 0: } \sqrt{\frac{2E_b}{T_b}} Cos(2\pi f_2 t + \theta_2)$$

   (i) Show that this BFSK signal may be expressed as the sum of two independent BASK signals.
   (ii) For θ_1 = (last two digits of your student ID)°
   and θ_2 = (middle two digits of your student ID)°, plot the BFSK waveform for the binary data stream "01101001" and compare your result to continuous-phase BFSK. (15)

3. (a) For each of the following four networks, discuss the consequences if a connection fails:
   (i) Six devices arranged in a bus topology
   (ii) Four devices arranged in a ring topology
   (iii) Five devices arranged in a mesh topology
   (iv) Seven devices arranged in a star topology (not counting the hub). (10)

   (b) What is the purpose of a reference model in data communication system? How is the OSI reference model different from the TCP/IP reference model? Mention the responsibilities of the each layer of the OSI model. (15)

4. (a) What is the number of bits per baud for the following techniques?
   (i) ASK with 8 different amplitudes
   (ii) FSK with 16 different frequencies
   (iii) PSK with different phases
   (iv) QAM with a constellation of 256 points. (12)

   (b) Draw the OQPSK signal components for the data "1100011001":
   (i) Modulating signal for in-phase component
   (ii) Modulated waveform of in-phase component
   (iii) Modulating signal for quadrature component
   (iv) Modulated waveform of quadrature component
   (v) OQPSK waveform. (13)

SECTION-B

5. (a) Explain chromatic dispersion and polarization mode dispersion in an optical fiber. How can such dispersion be compensated? (15)

   (b) A silica optical fiber has n_1 = 1.78 and n_2 = 1.76. Find (i) the critical propagation angle, (ii) the acceptance angle and (iii) the numerical aperture. How much will a light pulse spread after travelling along 6 km of this step-index fiber. (10)

6. (a) Compare the characteristics of LED and LD as optical source. (05)

   (b) Define quantum efficiency and responsitivity. Show that at 1248 nm wavelength, responsitivity and quantum efficiency is equal. (10)

   (c) When the mean optical power lunched into an 8km length of fiber is 125μW, the mean optical power at the fiber output is 4μW. Determine:

      (i) The overall signal attenuation or loss in decibels through the fiber assuming there are no connectors or splices;

      (ii) The signal attenuation per kilometer for the fiber;

      (iii) The overall signal attenuation for a 10 km optical link using the same fiber with splices at 1 km intervals, each giving an attenuation of 1 dB;

      (iv) The numerical input/output power ratio in (iii). (10)

7. (a) Define cell and cluster. Why is the shape of the cell chosen as hexagonal? (08)

   (b) Explain how the capacity of cellular system is enhanced by "cell-splitting" and "sectoring": (08)

   (c) What is meant by equalization and diversity techniques for communication systems? (09)

8. (a) A p-n photo diode has a quantum efficiency of 60% at a wavelength of 0.9μm. Calculate:

      (i) its responsitivity at 0.9μm

      (ii) the received optical power if mean photo current is $10^{-6}A$

      (iii) the corresponding number of received photons at this wavelength. (13)

   (b) With necessary diagram explain the working principle of continuous-wave (C.W.) radar. (12)

-THE END-

DHAKA UNIVERSITY OF ENGINEERING & TECHNOLOGY, GAZIPUR
B. Sc. Engineering 4th year 2nd Semester Examination, 2020
Department of Electrical and Electronic Engineering
Course Title: **Telecommunication-II**
Course No.: **EEE-4203**

Time: 3 hours | Full Marks: 210

Instructions: (a) Answer **any three questions** from each section.
(b) Use **separate script** for each section.
(c) Figures in the right margin indicate full marks.
(d) Necessary **graphs** should be provided.

SECTION-A

1. (a) Consider a linear modulation scheme for which the modulated wave is defined by equation (1). (12) CO1

$$s(t) = b(t)c(t) \quad (1)$$

where $b(t)$ is the incoming binary wave and $c(t)$ is the sinusoidal carrier which has unit energy measured over one-bit duration. Under the band-pass assumption, show that the transmitted signal energy (on a per bit basis) is a scaled version of the energy in $b(t)$.

(b) Explain the operation of the BFSK transmitter, and both the coherent and non-coherent BFSK receivers with necessary block diagrams. (13) CO1

(c) Fig. 1(c) shows a digital data transmission scheme. The baseband signal generator uses full-width pulses and polar signalling. The data rate is 1 Mbit/s. (10) CO2

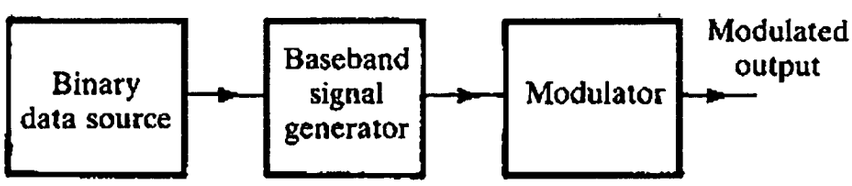


Fig. 1(c)

(i) If the modulator generates a 16-PSK signal, calculate the baud rate and bandwidth of the modulated output.
(ii) If the modulator generates a 4-FSK signal where the difference between the successive carrier frequencies is 100 kHz, determine the bandwidth of the modulated signal. For this 4-FSK case, assume that the transmission filter has a roll-off factor of 0.5.

2. (a) Explain the advantage of Gray encoding in multilevel modulation scheme with an example. Explain the relation between the bit error rate and the level of QAM scheme in noisy condition with the help of Fig. 2(a). Also, state the reason behind this sort of relationship. (10) CO1

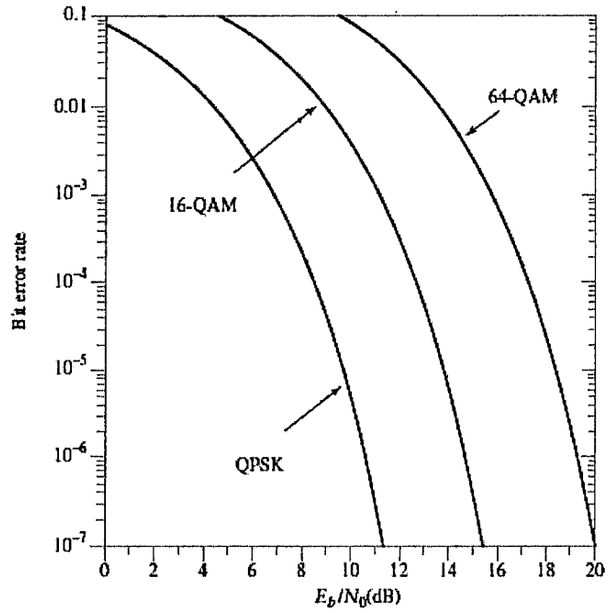
(b) Calculate the theoretical highest bit rate of a regular telephone line if the signal-to-noise ratio is 35 dB. A telephone line normally has a bandwidth of 3100 Hz (300 Hz to 3400 Hz). How many amplitude levels can be correctly detected in case of your calculated bit rate? (10) CO2

(c) Develop the radar range equation and comment on the relationship between the maximum range of radar and the operating frequency. (15) CO1

3. (a) What is the main function of transponder in a satellite communication system? Illustrate the block diagrams of the single- and double-frequency conversion transponders for 6/4 GHz and 14/11 GHz bands respectively. (12) CO1

(b) With a neat sketch, explain the main tasks of an earth station in a satellite communication system. (11) CO1

(c) Making necessary assumptions, sketch the wave shapes of the BASK, BPSK, BFSK and QPSK modulated signals to transmit a bit sequence of 0 1 1 0 1 1 0 0. (12) CO1



Comparison of BER versus E_b/N_0 for several digital transmission strategies.

Fig. 2(a)

4. (a) Explain simplex, half-duplex and full-duplex data transmission schemes with examples. Also, state the features of LANs, WANs and MANs. (10) CO1

(b) Explain the operation of the QPSK transmitter and the coherent QPSK receiver with necessary block diagrams. Also, sketch the constellation diagram for the QPSK transmitter you have drawn. (15) CO1

(c) Calculate the maximum range of radar for the following specifications: (10) CO1

Peak power transmitted by the radar = 400 kW
Operating frequency = 10 GHz
Effective aperture of the receiving antenna = 5 m^2
Radar cross section of the target = 30 m^2
Power of minimum detectable signal = 10^{-10} W

SECTION-B

5. (a) Define attenuation in an optical fiber. List three major causes of attenuation in an optical fiber and explain their mechanism. (12) CO1

(b) Define dispersion. Briefly explain chromatic dispersion and polarization mode dispersion phenomena. Also, describe their compensation technique. (13) CO1

(c) The mean optical power launched into an optical fiber link is 1.6 mW and the fiber has an attenuation of 0.26 dB/km. Determine the maximum possible link length without repeater when minimum optical power required at the detector is 2.5 μW. (10) CO2

6. (a) Explain the structure of a GSM network and describe each component with their role in the operation of GSM system. (13) CO1

(b) The velocity of light in the core of a step index fiber is 2.01×10^8 m/s and the critical angle at the core-cladding interface is 85°. Determine NA, acceptance angle and critical propagation angle. (11) CO2

(c) A p-n photodiode has a quantum efficiency of 60% at a wavelength of 0.9 μm. Calculate: (11) CO2
(i) its responsitivity.
(ii) the received optical power if photo current is 10 μA.
(iii) the corresponding number of received photons.

7. (a) Compare among various generations of cellular system. (12) CO1
(b) Differentiate among the TDMA, FDMA, and CDMA in wireless communication system. (11) CO1
(c) A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.51 and a cladding refractive index of 1.48. Determine: (12) CO2
(i) The critical propagation angles.
(ii) The critical angle at core cladding interface, and
(iii) The acceptance angle in air for the fiber.

8. (a) Mention the classification of handover. Briefly explain the handover process. (13) CO1
(b) Explain frequency reuse concept of a cellular network. Show that, for a hexagonal geometry, the co-channel reuse ratio $Q = \sqrt{3N}$. (10) CO1
(c) If a total of 33 MHz of bandwidth is allocated to a particular FDD cellular telephone system which uses a 50 kHz simplex channels to provide full duplex voice and control channels, compute the number of channels available per cell if a system uses (i) 4-cell reuse (ii) 7-cell reuse (iii) 12-cell reuse. If 1 MHz of the allocated spectrum is dedicated to control channels, determine an equitable distribution of control channels and voice channels in each cell for each of the three systems. (12) CO2

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